Repair the reasoning, not just the answer

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Coordinate sign / ES: signo de la coordenada / PT: sinal da coordenada

A minus sign is not "slow." Compare sizes for speed.

Also on your note page: Distance vs. displacement, Net force, Interaction pair.

B READ THE RELATIONSHIP

A sign shows direction on the axis you chose. It can change the meaning of an answer.

Find the first wrong step, repair it, then recheck

C TRACE THE ORIGINAL FIGURE

1 **CLAIM** ✗ "3 m right then 3 m back – the distance is zero."	1 **CLAIM** ✗ "8 N pull on a 2 kg cart: 8 ÷ 2 = 4 m/s²."
2 **FIX** ✓ Distance = 3 + 3 = 6 m. *Displacement* is 0 m: the endpoints match.	2 **FIX** ✓ Net force first: 8 – 2 = 6 N. Then 6 ÷ 2 = 3 m/s².
1 **CLAIM** ✗ "–2 → –6 m/s is slowing, because it's negative."	1 **CLAIM** ✗ "The force pair cancels, so nothing moves."
2 **FIX** ✓ Speed grows from 2 to 6 m/s: speeding up. The minus sign is direction.	2 **FIX** ✓ Each pair force acts on a different cart. Draw one on each diagram.
☐ **1 · PENCIL** Mistaken claims: cross out	☐ **2 · CODE COLOR** The fixes: color by quantity

Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg cart has 9 N right and 3 N left. A student writes 9 ÷ 3 = 3 m/s². Correct it.

Given m = 3 kg 9 N right 3 N left

Rule a = F net ÷ m

Work F net = 9 – 3 = 6 N right a = 6 ÷ 3

Answer 2 m/s² right missed: net force first

E SAVE FOR THE FINAL MEETING

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

Read the physics: Coordinate sign

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Write the positive direction first. Find the first step where the model, sign or starting value went wrong. Changing only the final number can hide the mistake.

02 HOW IT WORKS

When forces oppose, give them opposite signs before adding. When reading velocity, compare its size to tell whether speed grows. A more negative velocity can mean a larger speed.

03 WHAT TO CHECK

Repair the first wrong step. Then use a sketch, force sum or unit check to test the new reasoning.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg cart has 9 N right and 3 N left. A student writes $9 \div 3 = 3$ m/s². Correct it. 2 m/s² right missed: net force first.

05 CONNECT TO XJ

A forecast is only as good as its reasoning. Before the desk signs off on XJ, every step is checked: signs, starting position, units, and the right model.

SAY IT BEFORE YOU WRITE

What failed first: the rule, the sign or the arithmetic? The first error was ___. I replaced it with ___ because ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Coordinate sign / ES: signo de la coordenada / PT: sinal da coordenada

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, coordinate sign means …	You choose right as positive. What sign goes on a leftward force? A leftward force gets a ___ sign on this axis.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

Labels you can use: first error · repair · check.

First wrong step	Repair

Tell a partner: The first error was ___. I replaced it with ___ because ___.

On assignment: Coordinate sign

SECTION E / BACK / FINAL MEETING ONLY Name _____

A calculator overlay for a 3 kg tracker shows +9 N and −3 N. Its output is "9 ÷ 3 = 3 m/s².” It is about to go live.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Software tester**

File a bug ticket: mark the first wrong step, enter the corrected force sum and give the expected display.

☐ **Fact-check producer**

Stop or clear the graphic. Give the anchor a corrected number and a short reason viewers can check on the force arrows.

Word bank: first error · repair · check.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”